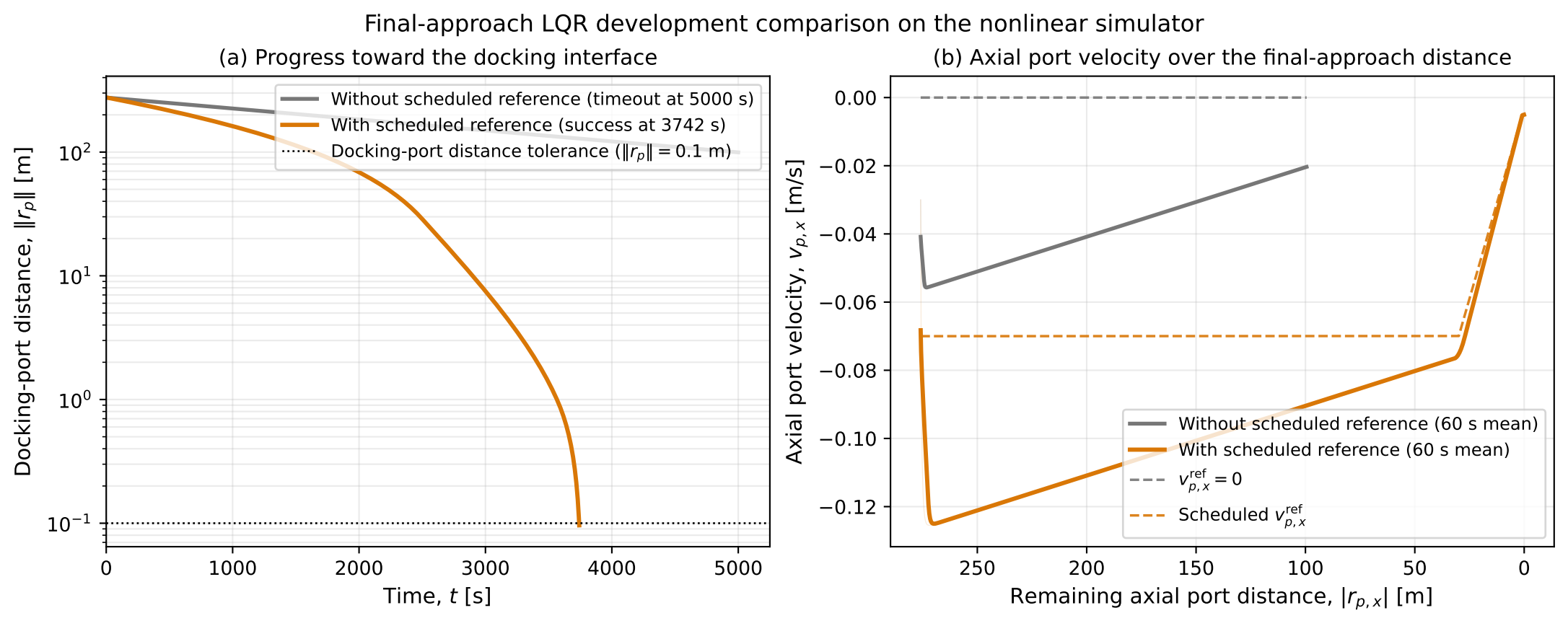
Final-approach LQR development comparison on the nonlinear simulator
(a) Progress toward the docking interface
(b) Axial port velocity over the final-approach distance
Without scheduled reference (timeout at 5000 s)
With scheduled reference (success at 3742 s)
Docking-port distance tolerance ($\|r_p\| = 0.1$ m)
Docking-port distance, $\|r_p\|$ [m]
Time, t [s]
Axial port velocity, $v_{p,x}$ [m/s]
Remaining axial port distance, $|r_{p,x}|$ [m]
Without scheduled reference (60 s mean)
With scheduled reference (60 s mean)
$v_{p,x}^{\mathrm{ref}} = 0$
Scheduled $v_{p,x}^{\mathrm{ref}}$